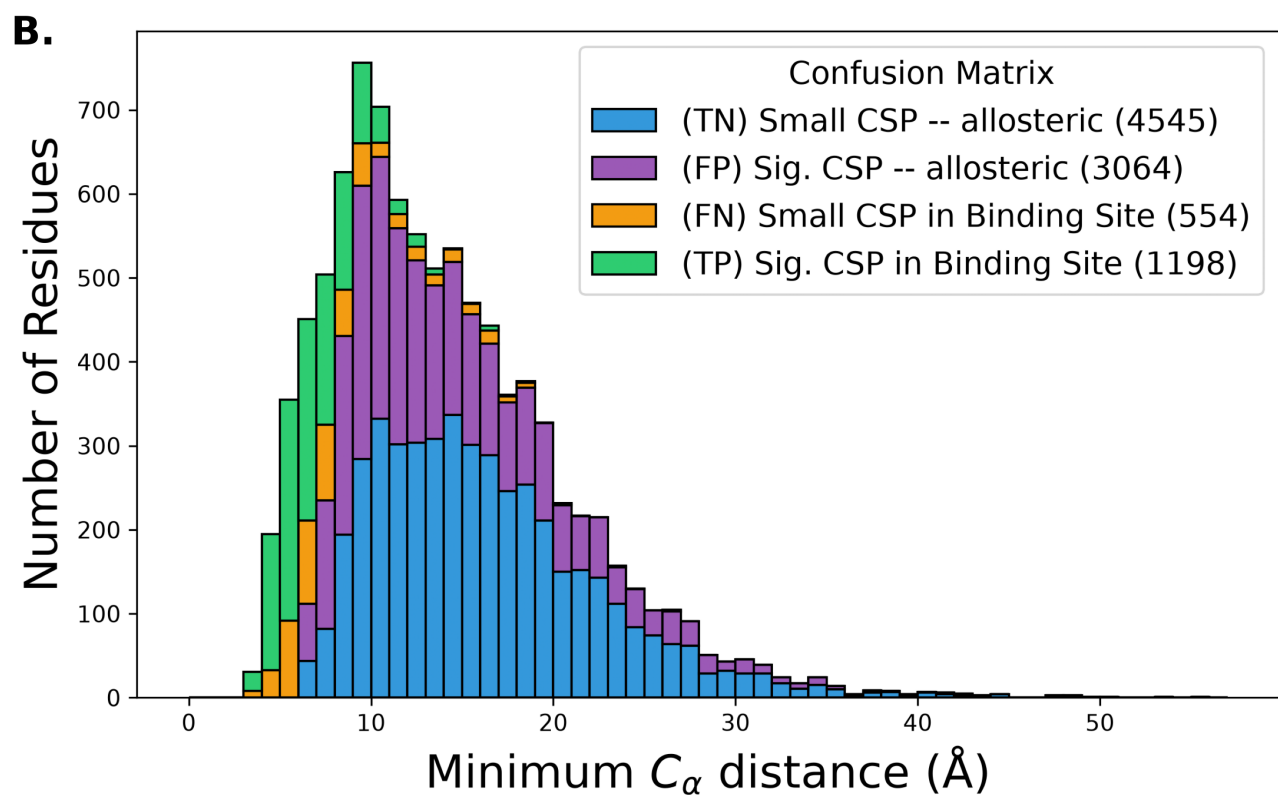
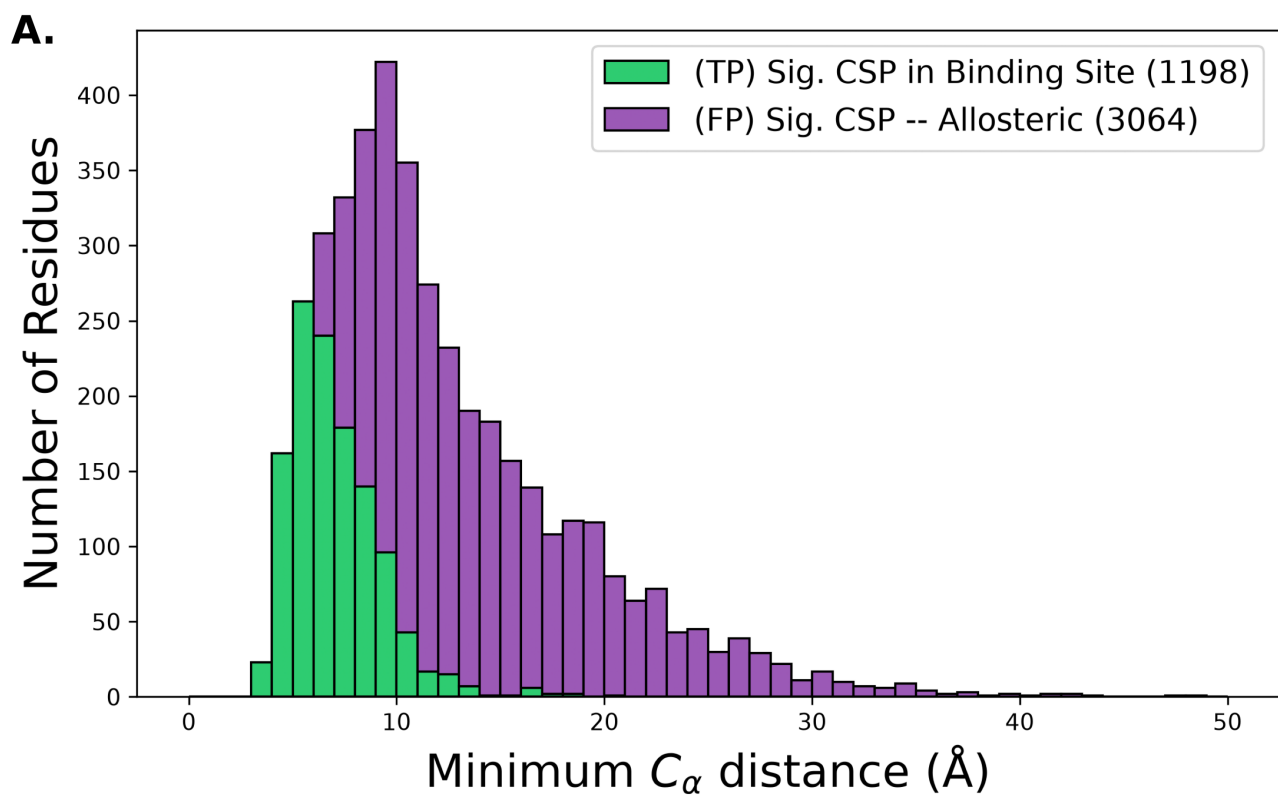
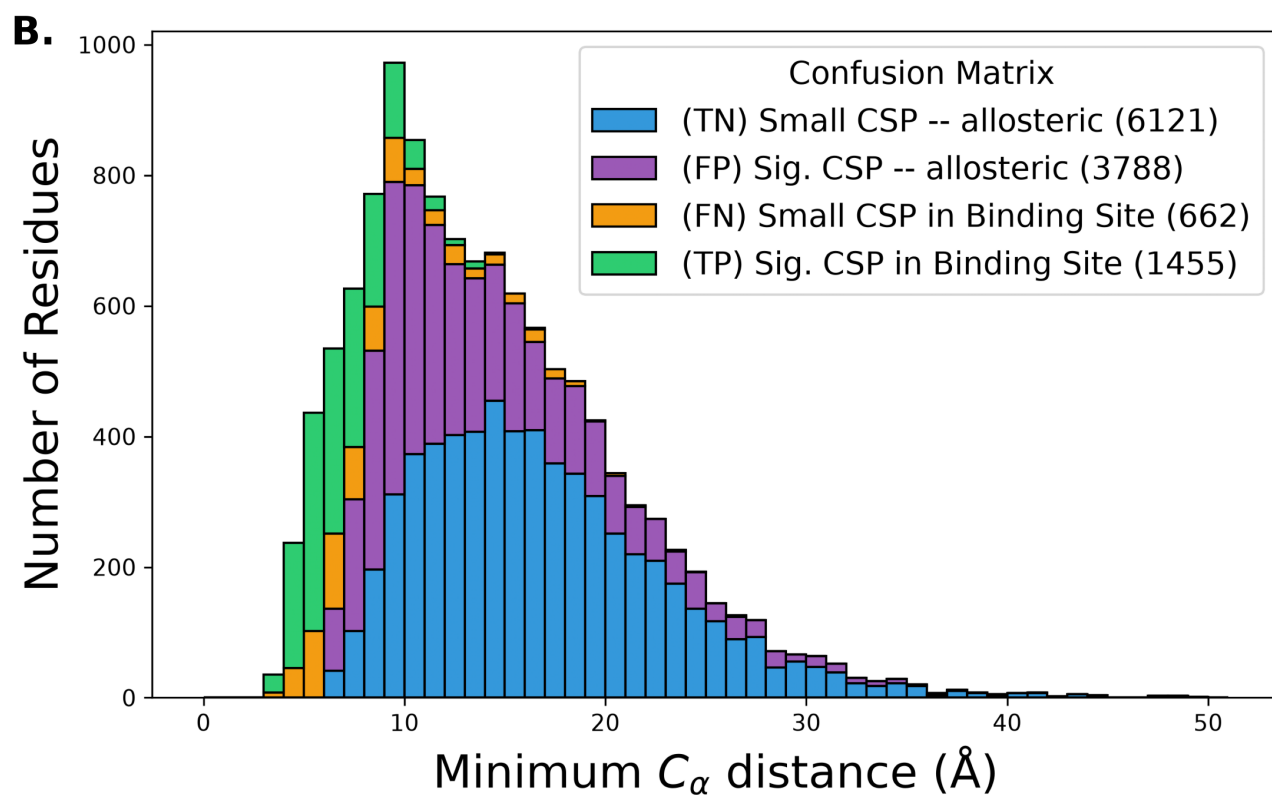
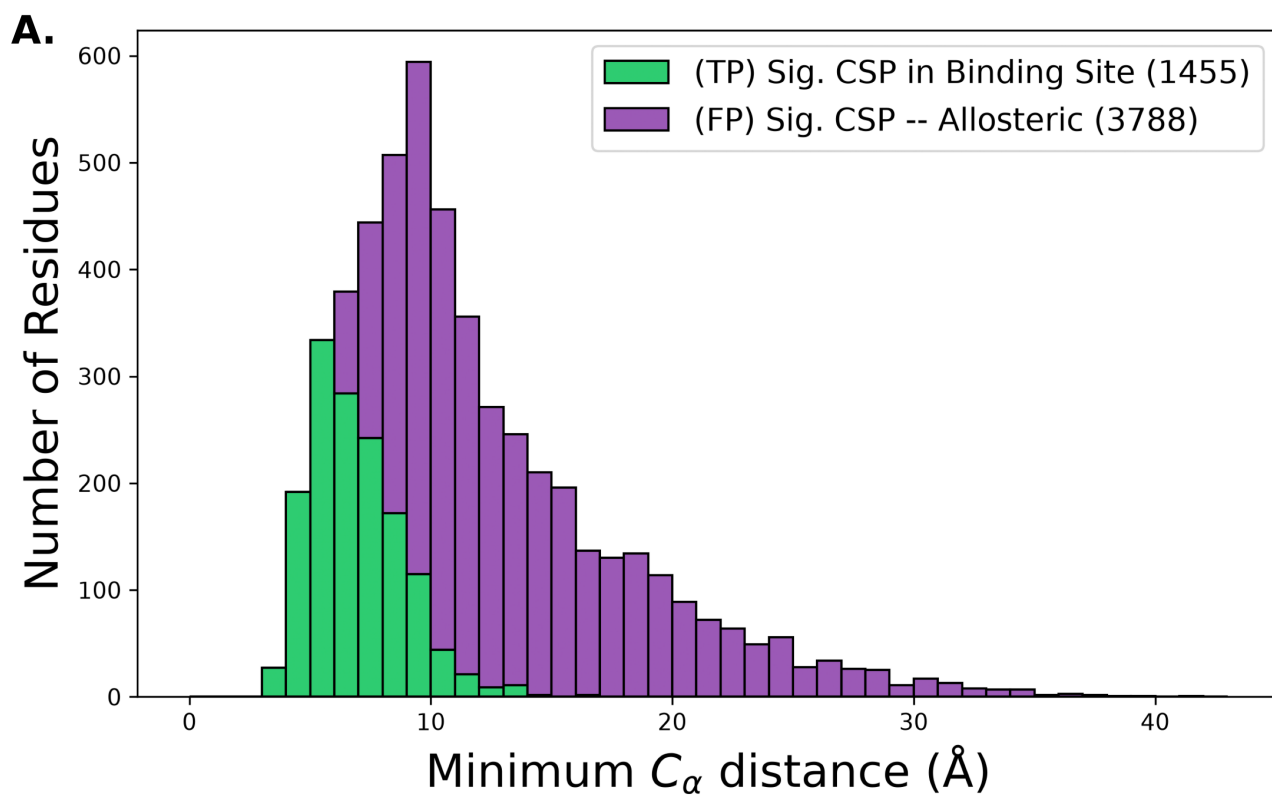
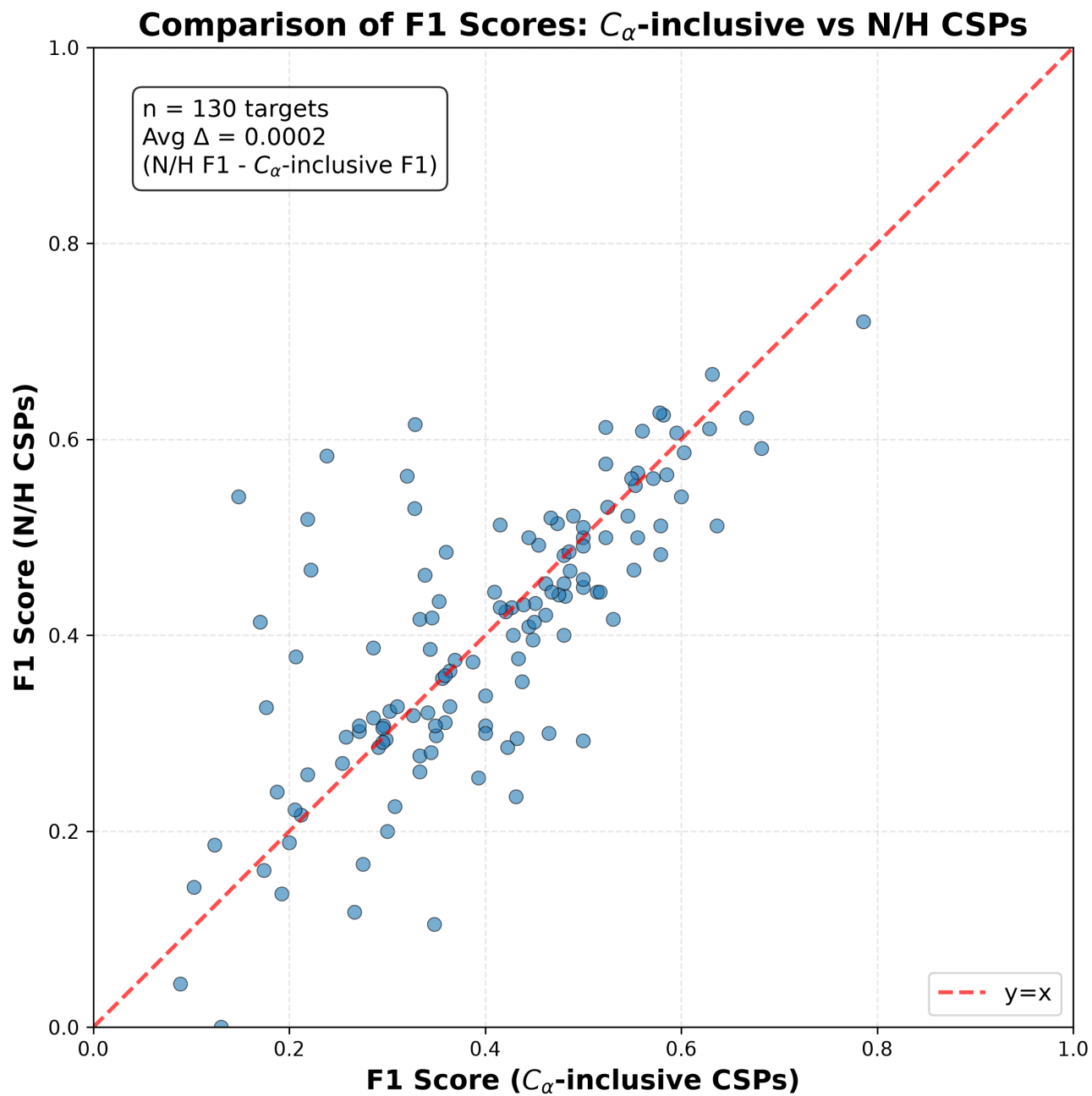




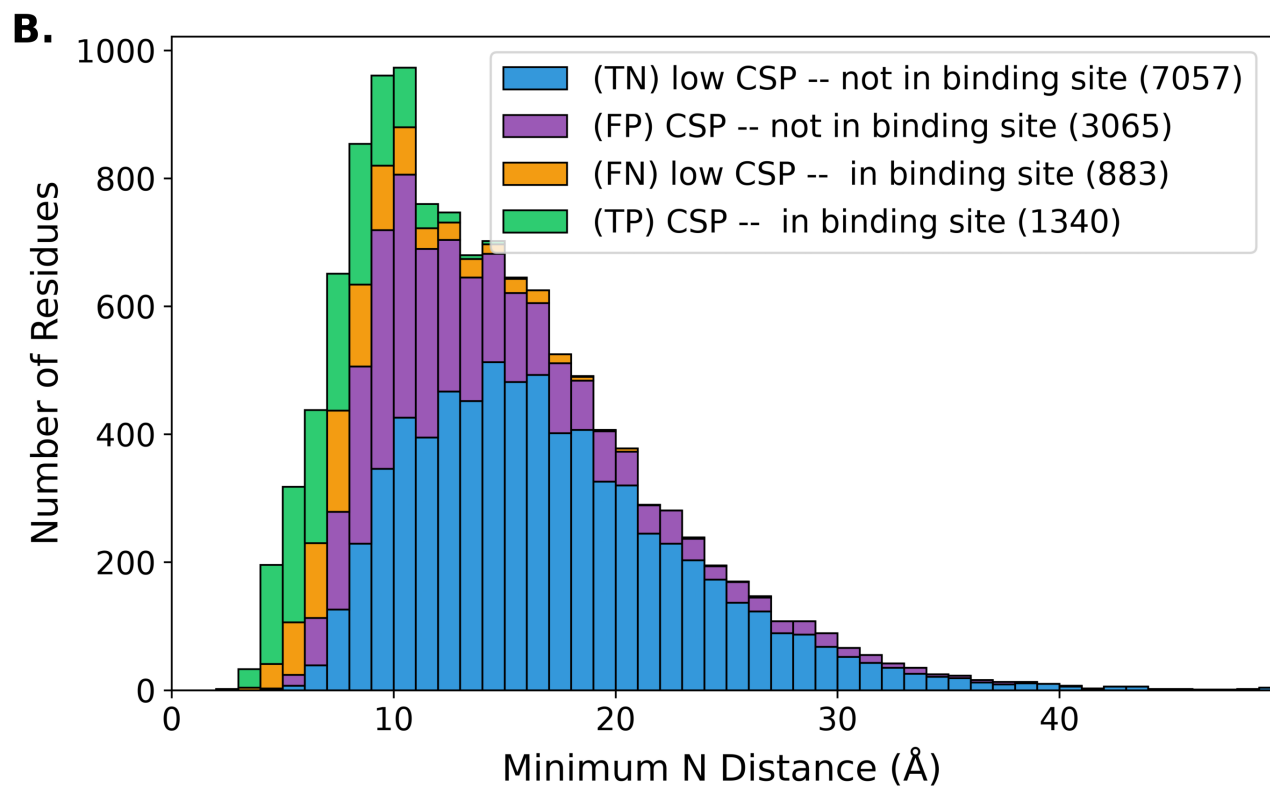
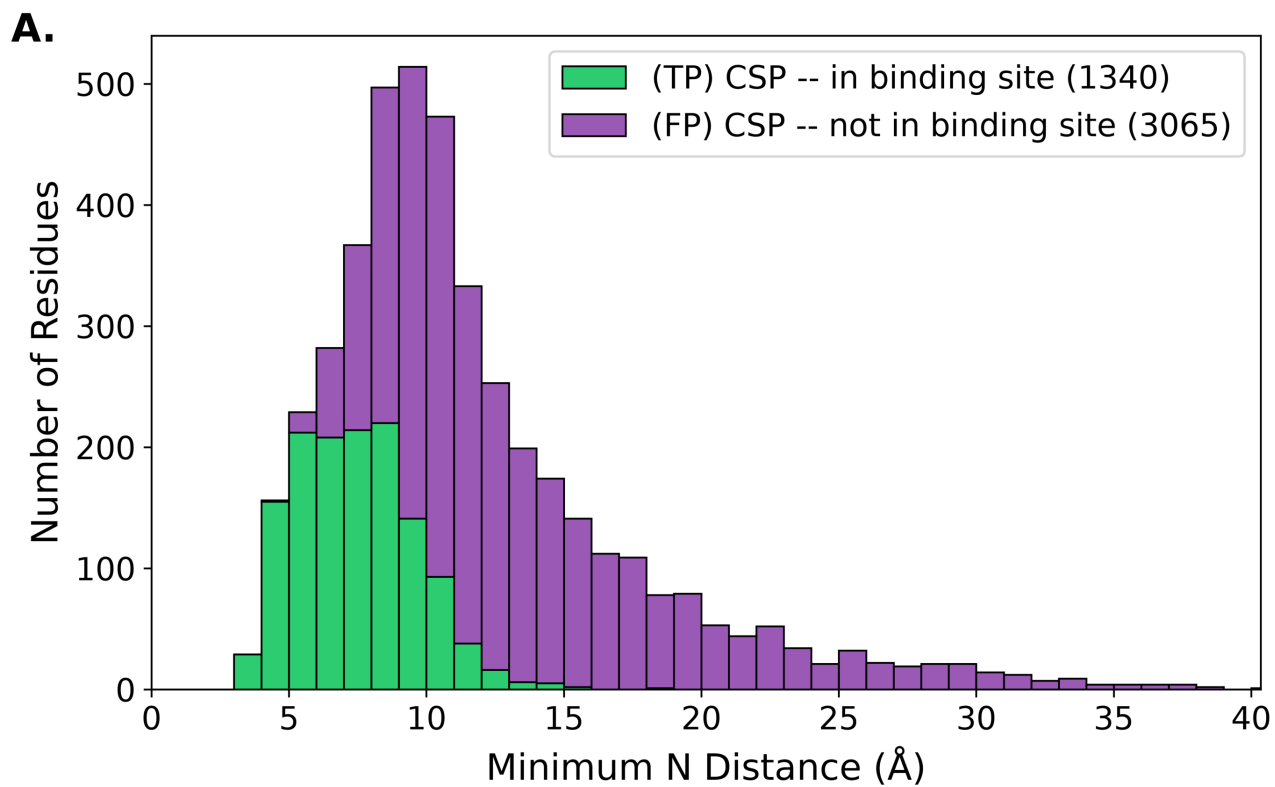
SI Fig. S10. HA/CA CSP confusion histograms for the buffer-similar systems from CSP_UBQ_ph0.5_temp5C.csv that also have H_{α} and C_{α} shifts in both apo and holo experiments ($n = 101$).



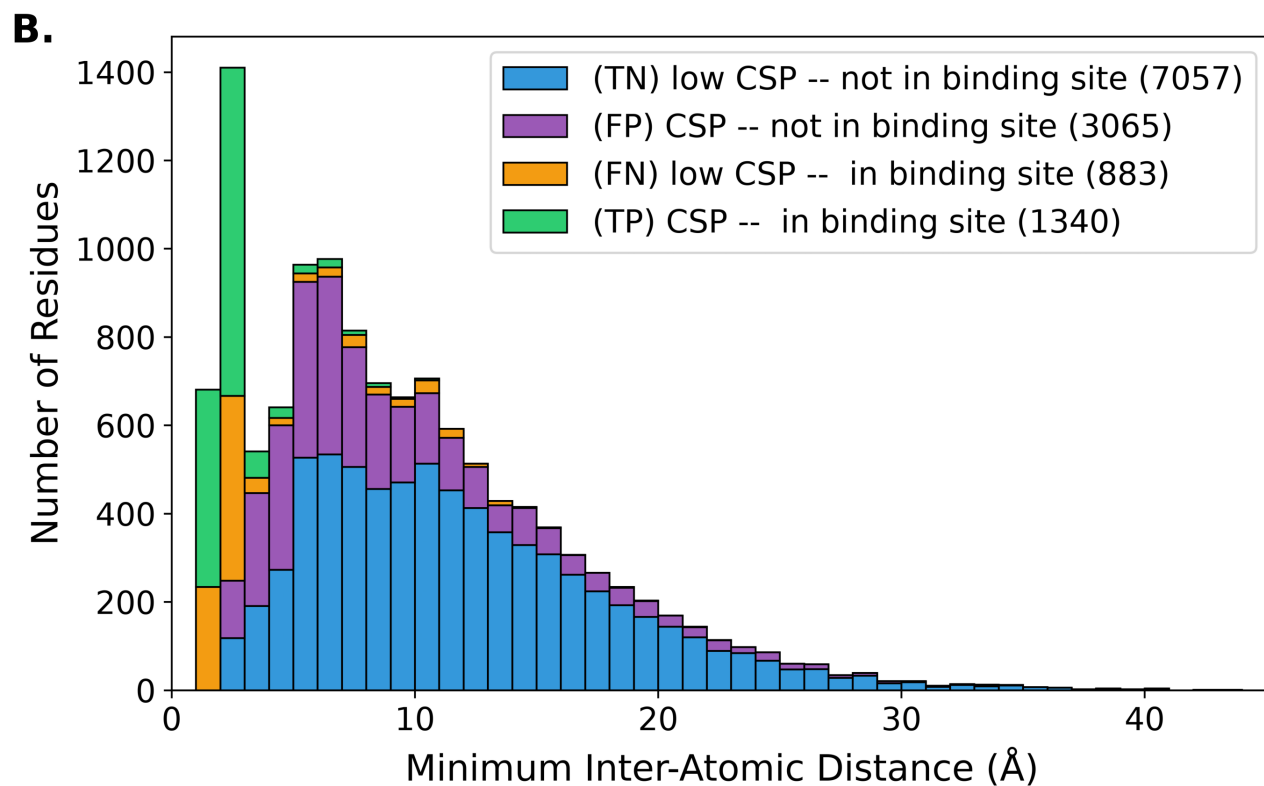
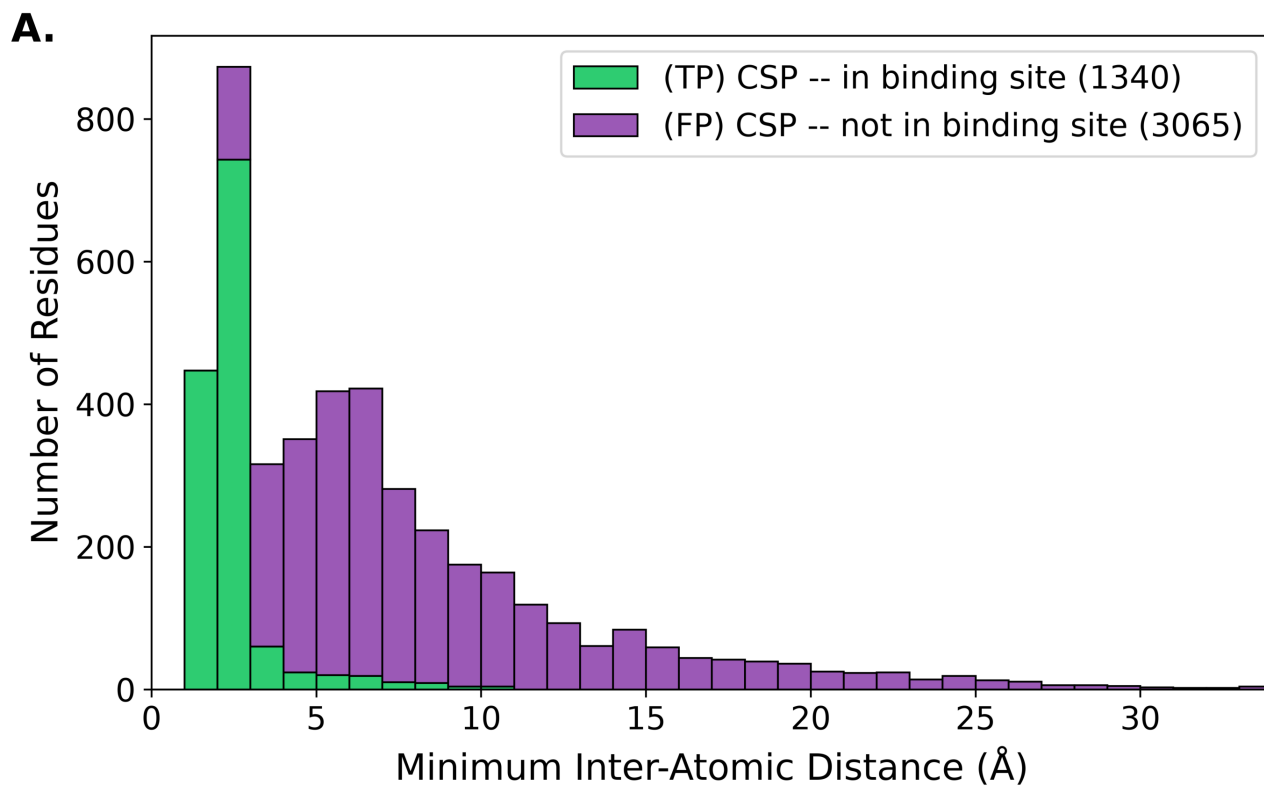
SI Fig. S11. CA-inclusive CSP confusion histograms for the $n = 145$ buffer-similar systems from CSP_UBQ_ph0.5_temp5C.csv.



SI Fig. S12. F1 scores for CA-inclusive vs exclusive CSPs ($n = 145$ buffer-similar systems from CSP_UBQ_ph0.5_temp5C.csv).

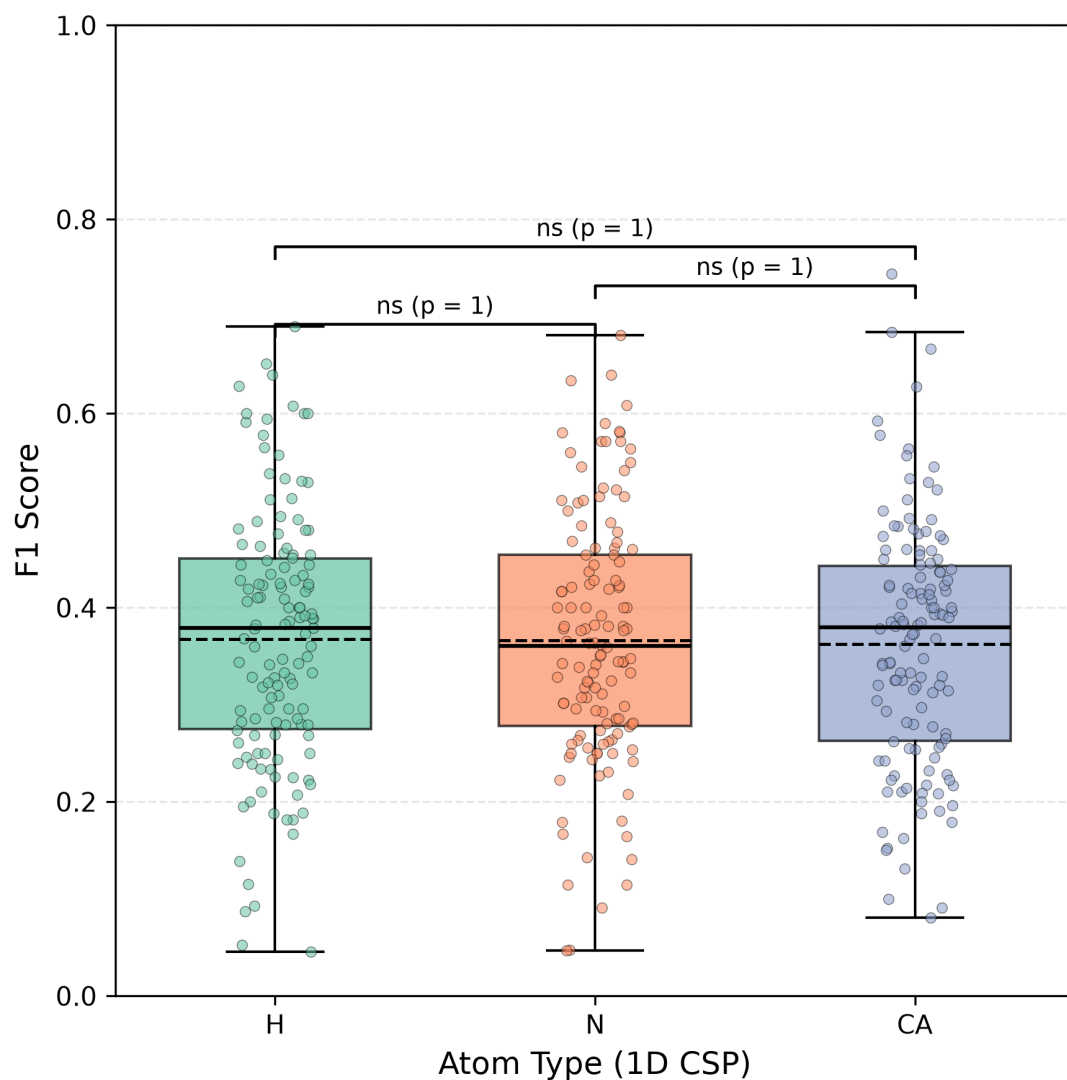


SI Fig. S13. Confusion histograms by closest interchain N–N distance ($n = 136$ original pH/temp-matched systems, excluding rematch pairs).

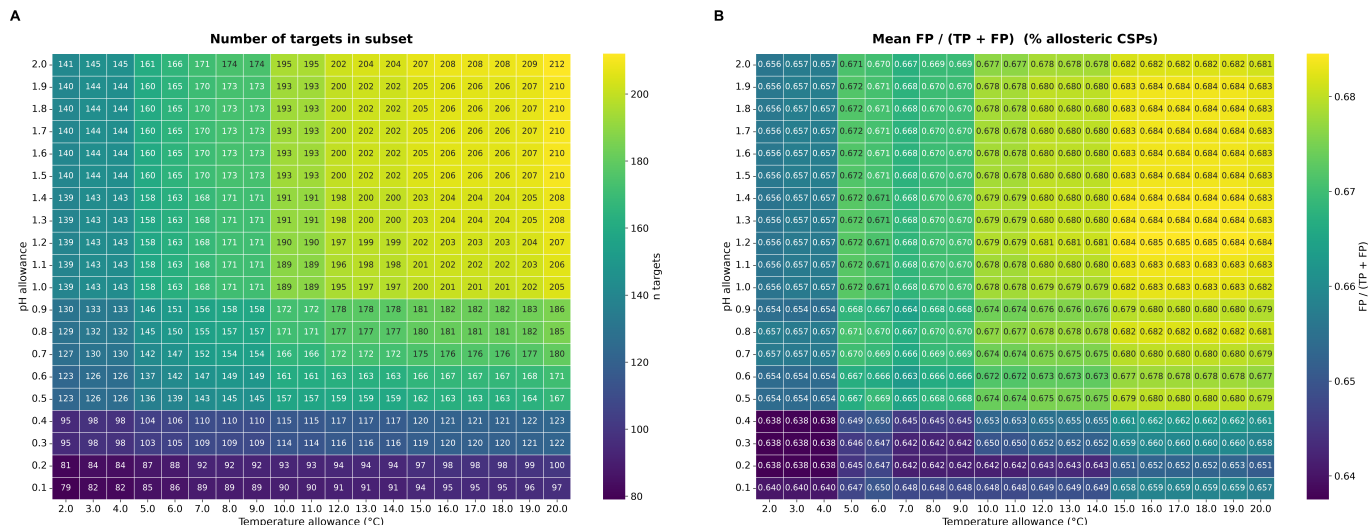


SI Fig. S14. Confusion histograms by closest interchain any-atom distance ($n = 136$ original pH/temp-matched systems, excluding rematch pairs).

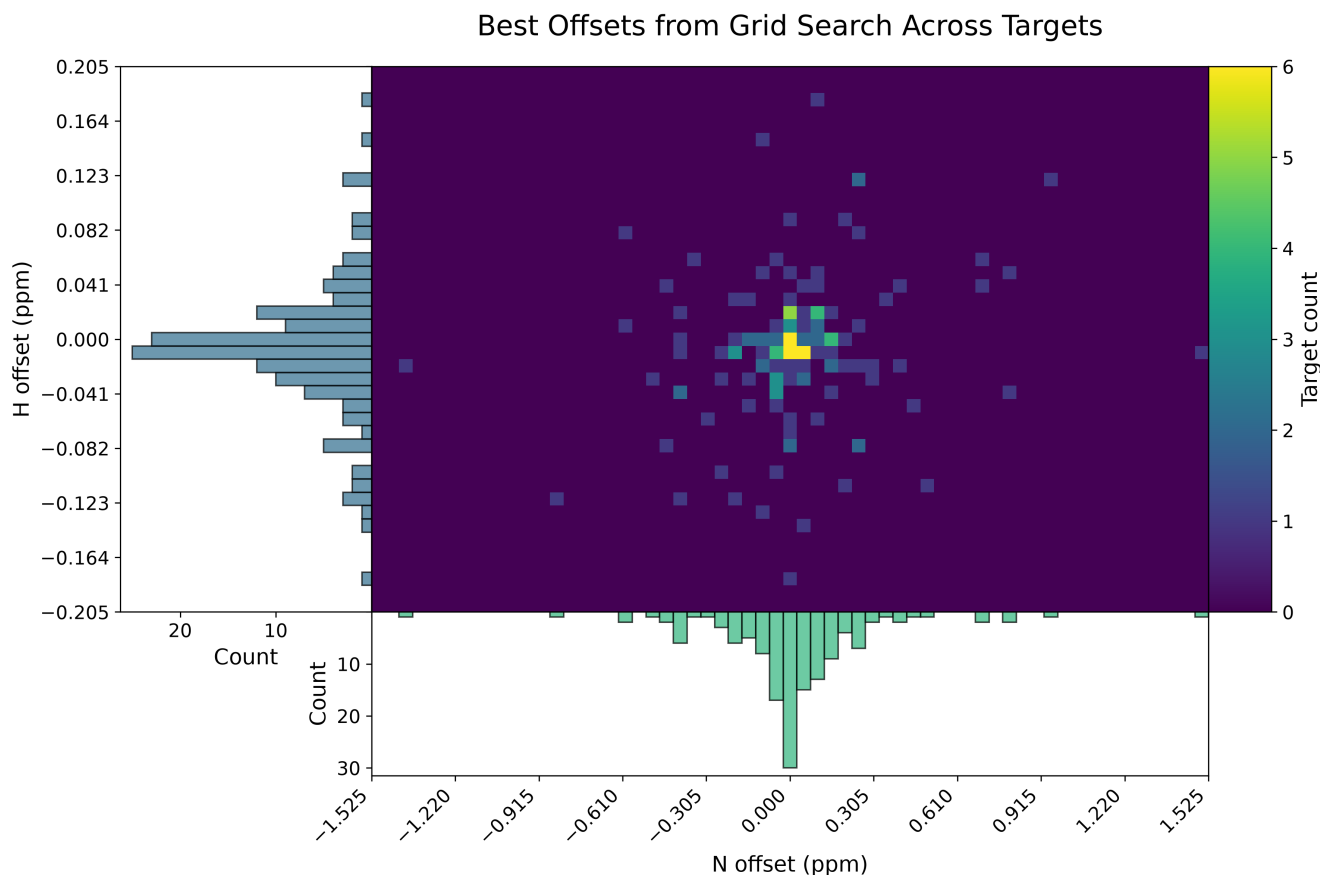
F1 score distributions for 1D single-atom CSPs (n = 130 targets)



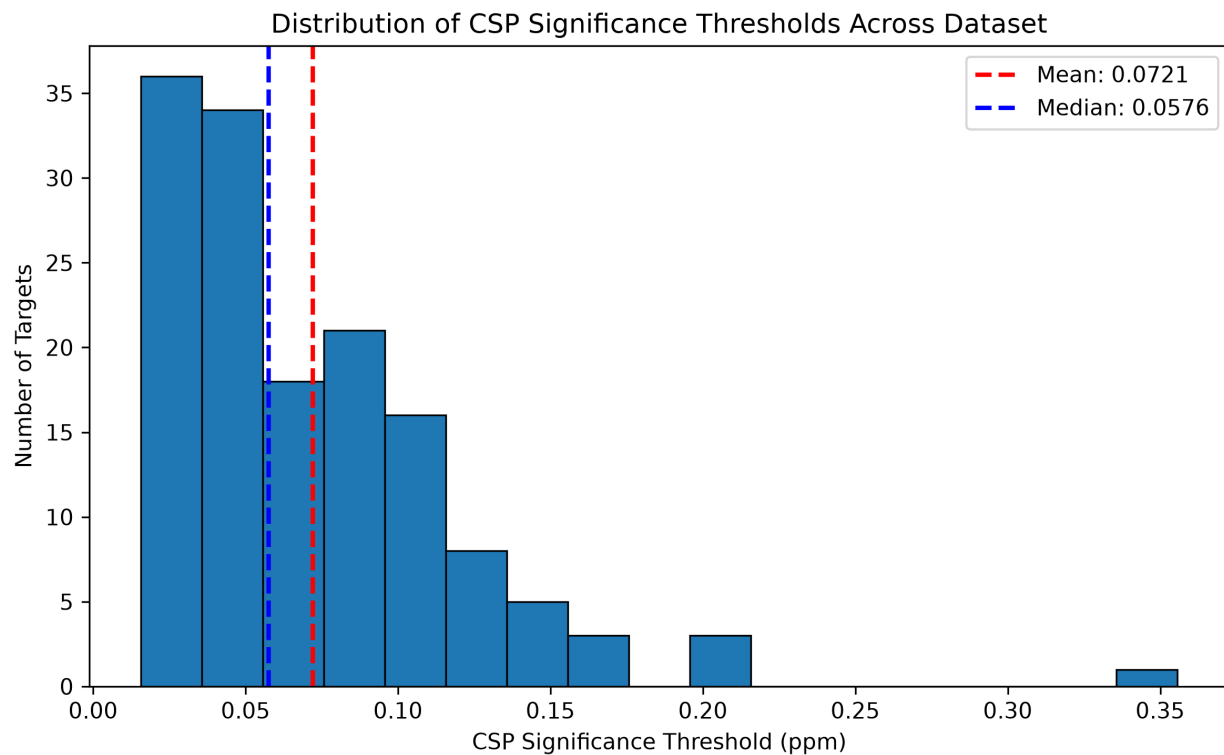
SI Fig. S15. Per-target F1 scores for 1D H/N/C α CSPs ($n = 145$ buffer-similar systems from CSP_UBQ_ph0.5_temp5C.csv).



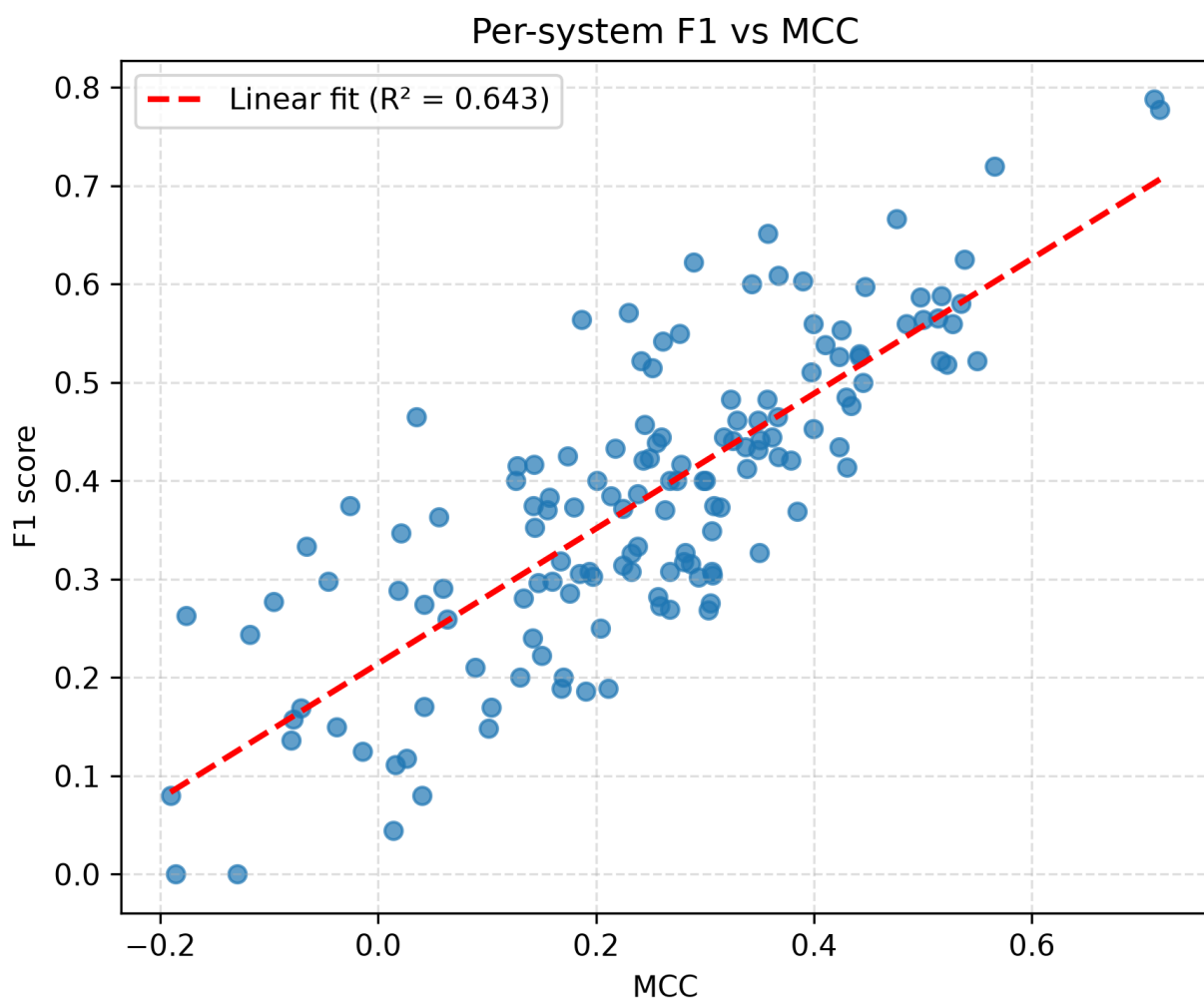
SI Fig. S17. Buffer pH/temperature threshold sweep



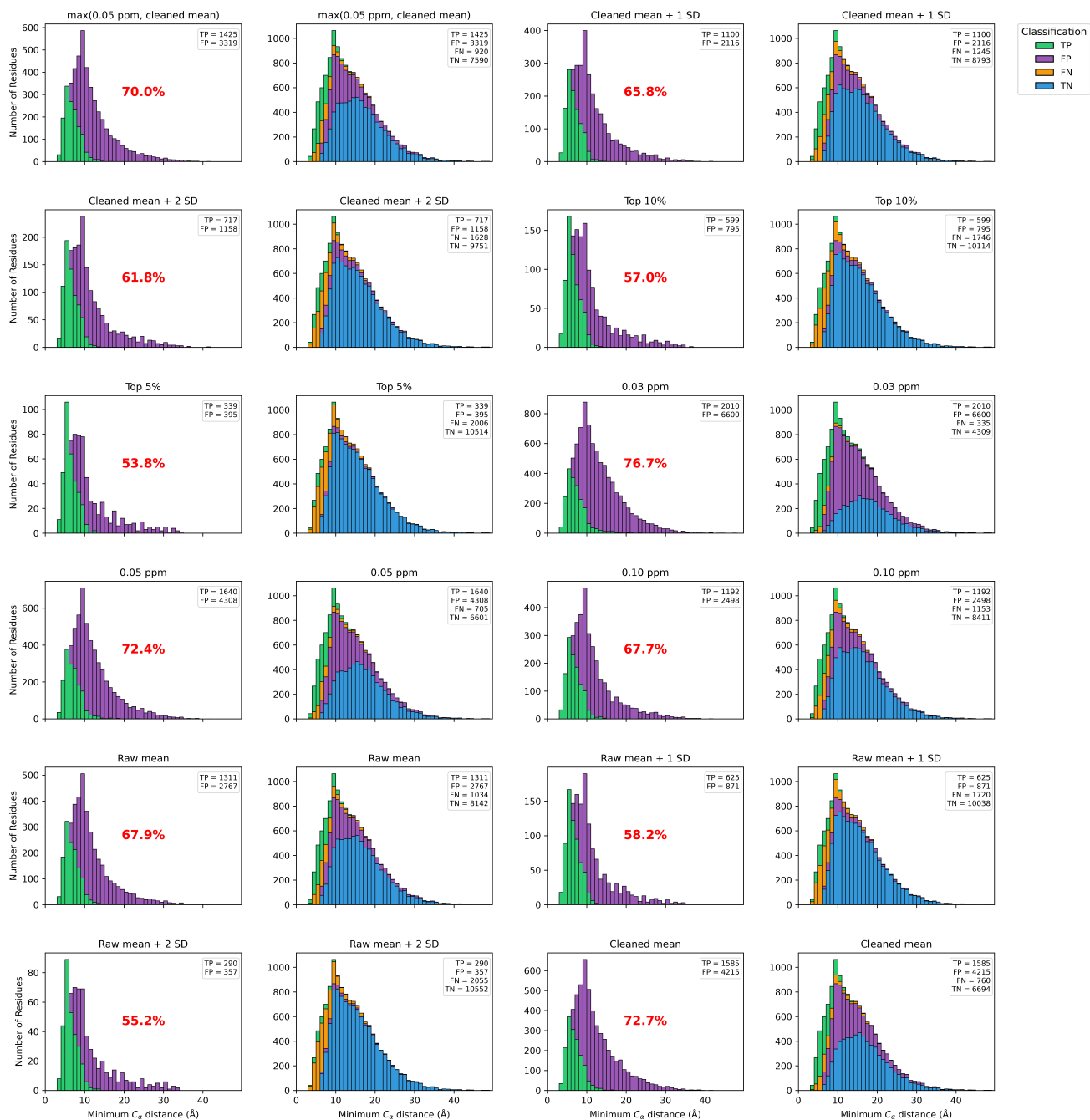
SI Fig. S18. Ideal $^1\text{H}/^{15}\text{N}$ chemical-shift offset grid for the $n = 145$ buffer-similar systems in CSP_UBQ_ph0.5_temp5C.csv. For each target, a grid search selects the H/N referencing offsets that maximize aligned HN peak counts under the pipeline CSP criterion; the heatmap aggregates those optimal $(\Delta\delta_{\text{N}}, \Delta\delta_{\text{H}})$ pairs (color = number of targets). Marginal histograms show the one-dimensional distributions of best H (left) and N (bottom) offsets. Both are peaked near zero, indicating that most apo/holo pairs require only small referencing corrections.



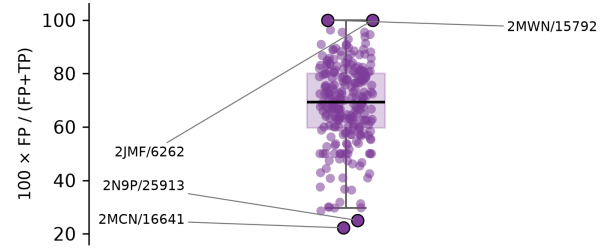
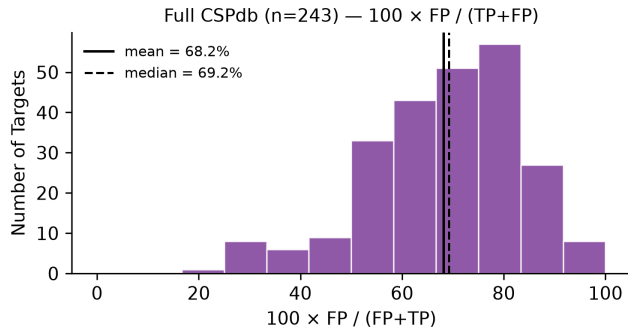
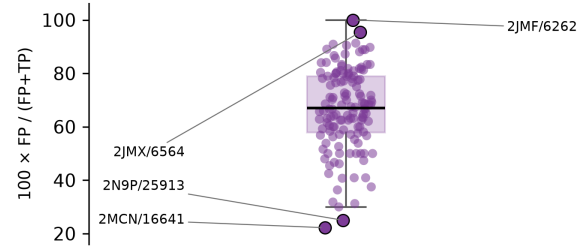
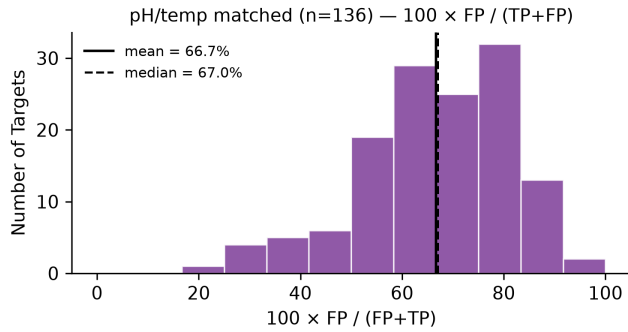
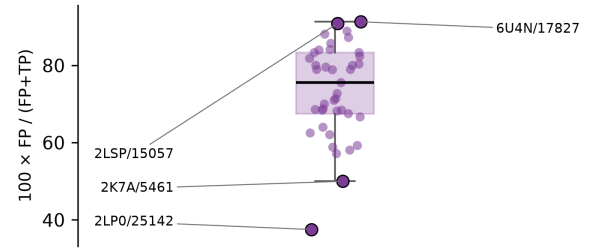
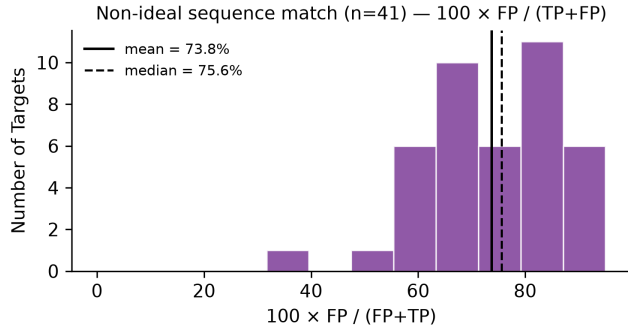
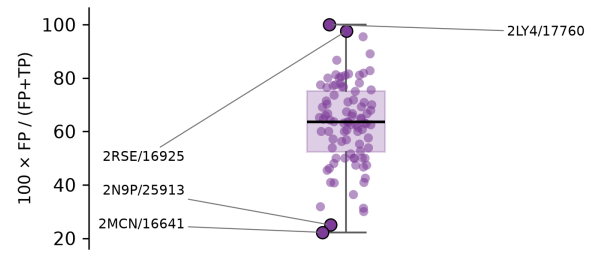
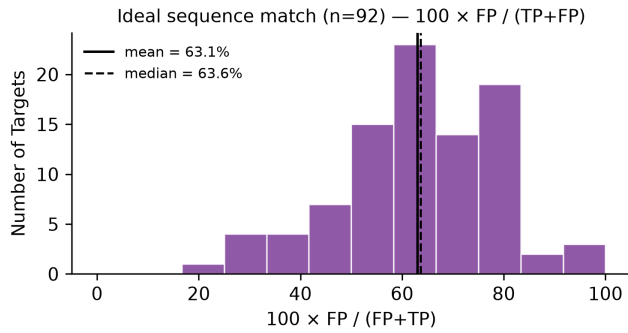
SI Fig. S19. Distribution of CSP significance thresholds ($n = 145$ buffer-similar systems from CSP_UBQ_ph0.5_temp5C.csv).



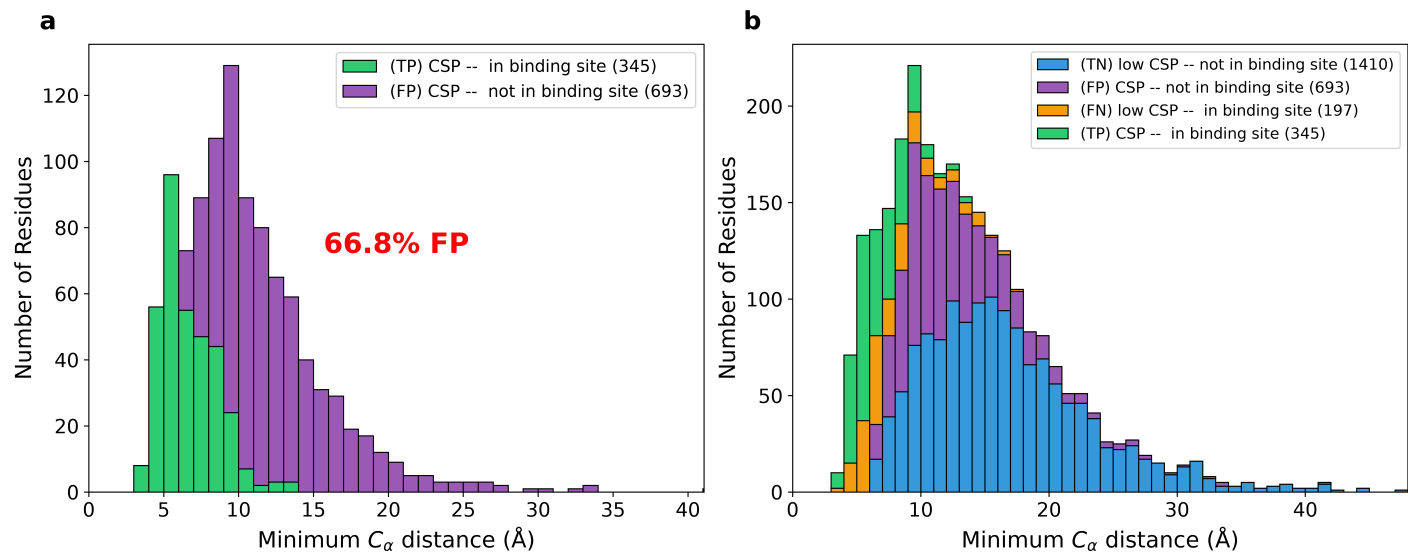
SI Fig. S20. F1 vs MCC across the $n = 145$ buffer-similar systems in `CSP_UBQ_ph0.5_temp5C.csv`.



SI Fig. S21. Figure 3-style CA-distance histograms under alternate HN CSP significance thresholds ($n = 145$ buffer-similar systems from CSP_UBQ_ph0.5_temp5C.csv). Each column is a 3a (TP/FP among significant CSPs) | 3b (full confusion) pair. Thresholds (left to right, then continuing as labeled on the figure): **primary** max(cleaned mean, 0.05 ppm) after iterative outlier removal; **cleaned mean** +1 SD and +2 SD; **top 10%** and **top 5%** of HN CSP values; fixed cutoffs **0.03**, **0.05**, and **0.10 ppm**; **raw mean**, **raw mean** +1 SD, and **raw mean** +2 SD (no outlier removal); and **cleaned mean** without the 0.05 ppm floor.

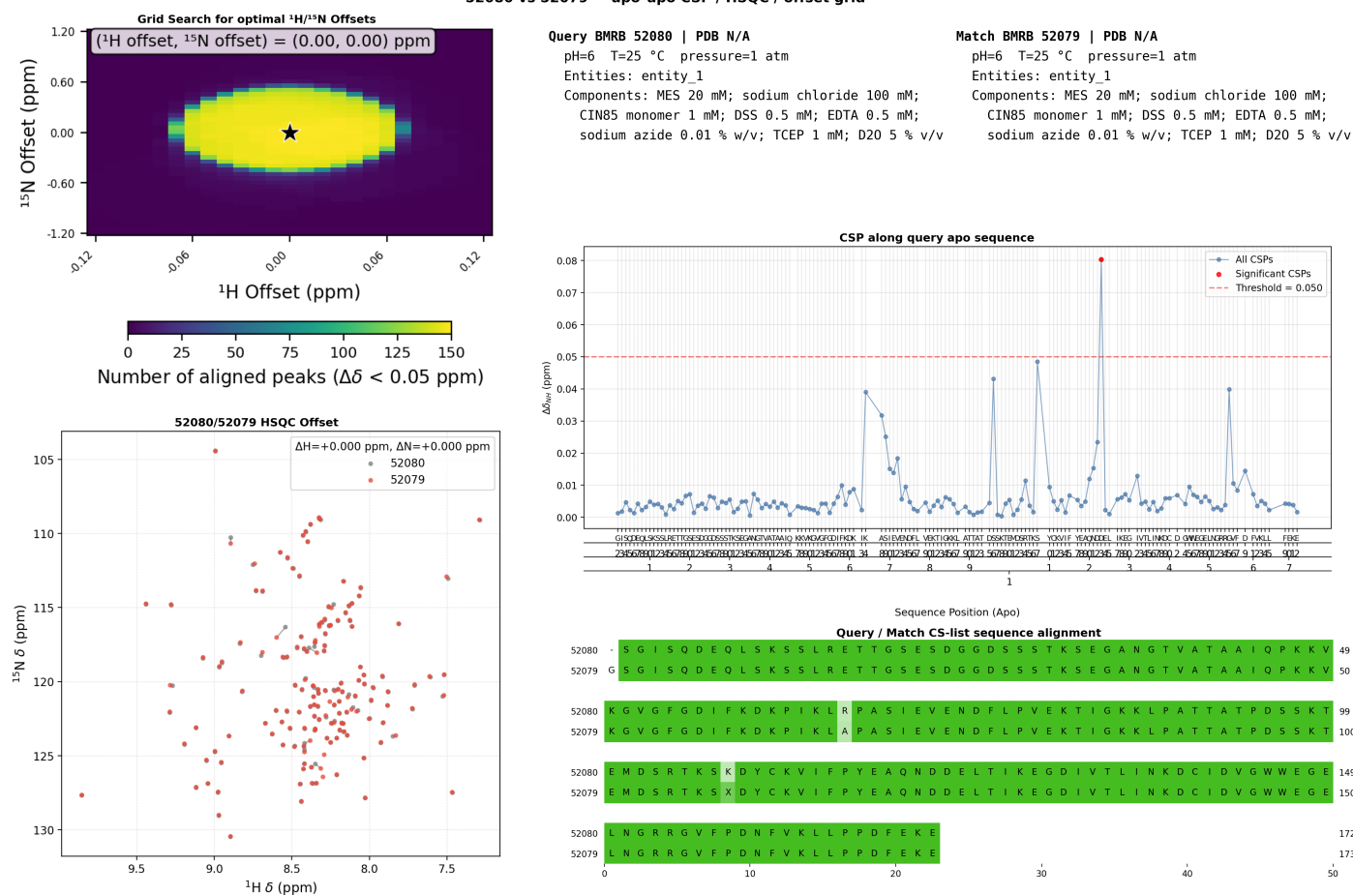


SI Fig. S22. Per-target false-positive rates under the primary CSP cutoff max(cleaned mean, 0.05 ppm). Rows show four cohorts: **ideal sequence match** ($n = 92$), **non-ideal sequence match** ($n = 41$), **pH/temp matched** ($n = 136$), and the **full CSPdb** ($n = 244$). For each cohort, panels report $100 \times \text{FP} / (\text{TP} + \text{FP})$ among significant CSPs.



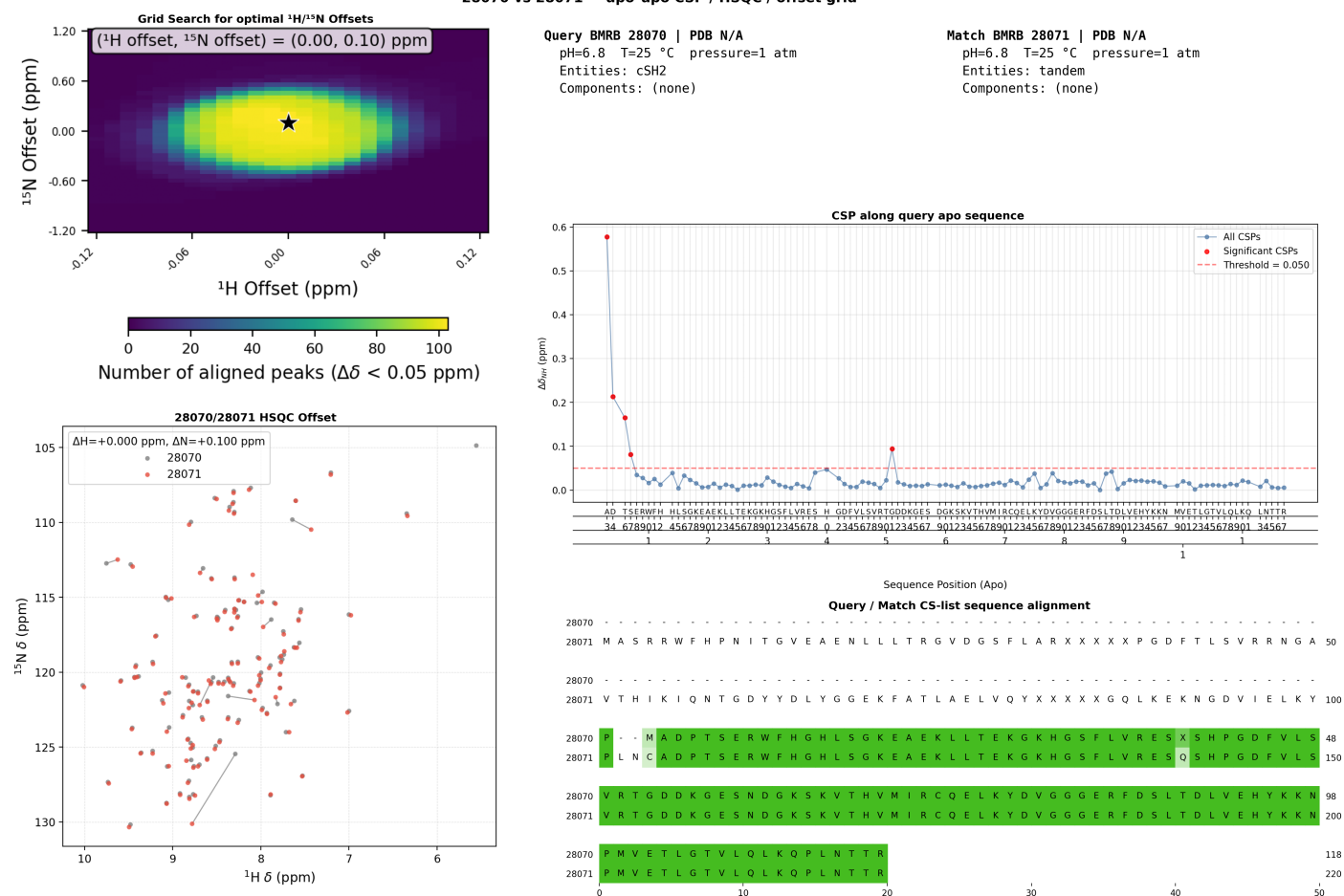
SI Fig. S23. Figure 3 histograms for the same-author/same-study subset. Targets are the curated pairs in `CSP_UBQ_ph0.5_temp5C_same_author_list.csv` ($n = 33$): buffer-similar apo/holo systems that share a study or last author between apo and holo chemical-shift depositions. Left: significant-CSP TP vs FP by minimum C_α distance to the interface; right: full confusion-matrix stacks (TN/FP/FN/TP).

52080 vs 52079 — apo-apo CSP / HSQC / offset grid



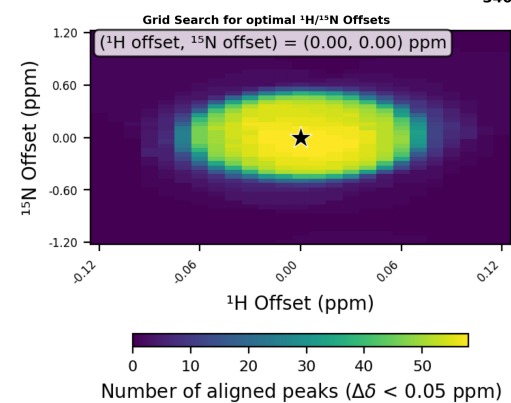
SI Fig. S24A. Apo vs apo control study for BMRB 52080 and 52079. This case involves the CIN85 SH3B–SH3C linker (residues 163–333). There is a sequence mismatch at R227A (BMRB 52080 is the R229A variant; BMRB 52079 is R227A/R229A); BMRB 52079 also has an extra N-terminal Gly. 99.3% of the CSPs for this control are below 0.05 ppm.

28070 vs 28071 — apo-apo CSP / HSQC / offset grid



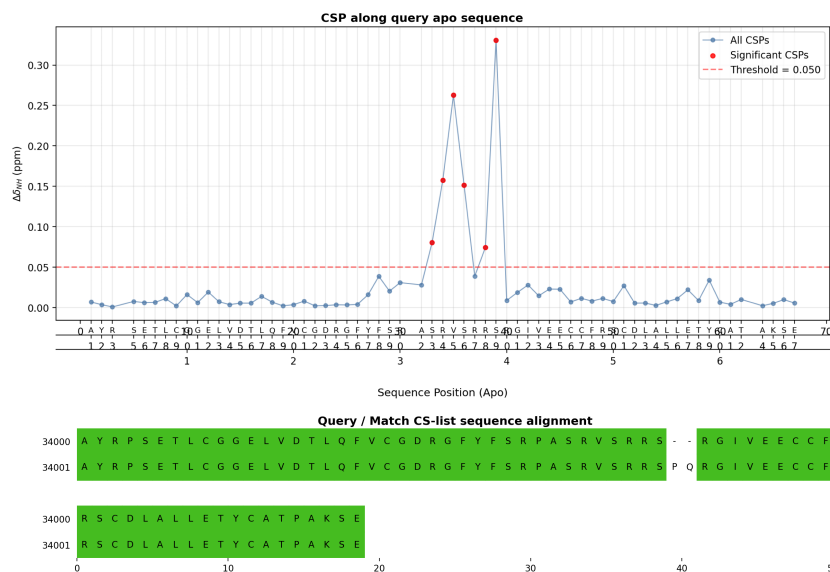
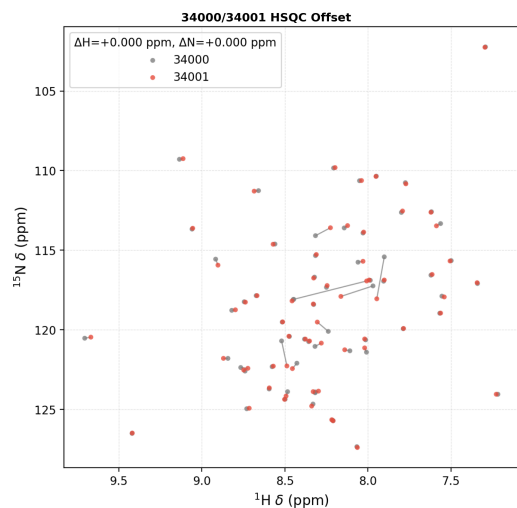
SI Fig. S24B. Apo vs apo control study for BMRB 28070 and 28071. This case involves the SH2 domains of SHP2. BMRB 28071 is the tandem nSH2–cSH2 construct and includes a large N-terminal region (the nSH2 domain and inter-SH2 linker, ~100 residues) that is absent from the isolated cSH2 entry (BMRB 28070). 95.4% of the CSPs for this control are below 0.05 ppm.

34000 vs 34001 — apo-apo CSP / HSQC / offset grid



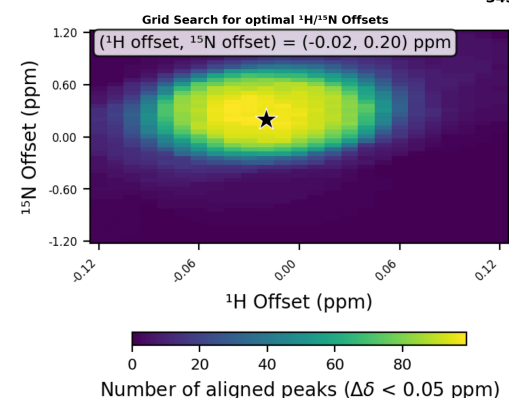
Query BMRB 34000 | PDB N/A
pH=4.2 T=35 °C pressure=1 atm
Entities: Insulin-like growth factor II
Components: (none)

Match BMRB 34001 | PDB N/A
pH=4.2 T=35 °C pressure=1 atm
Entities: Insulin-like growth factor II
Components: (none)



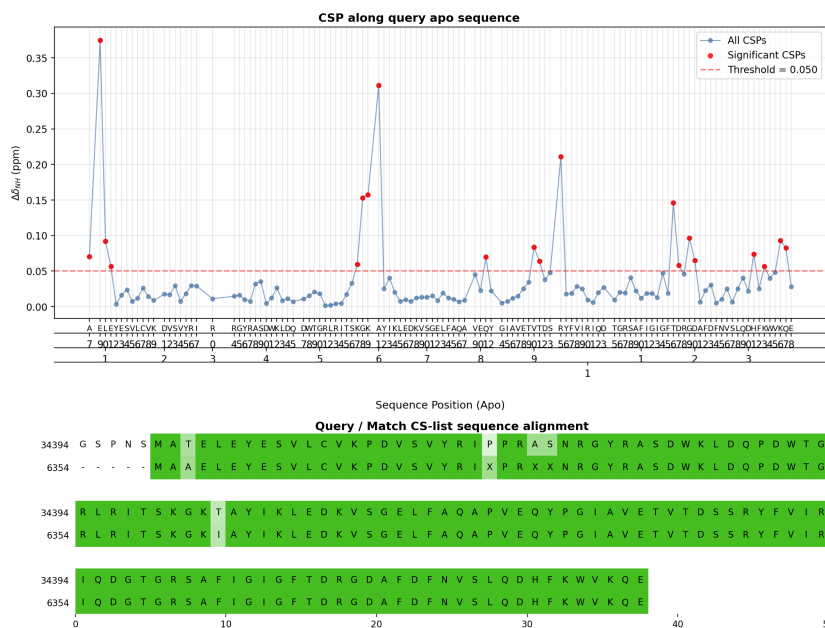
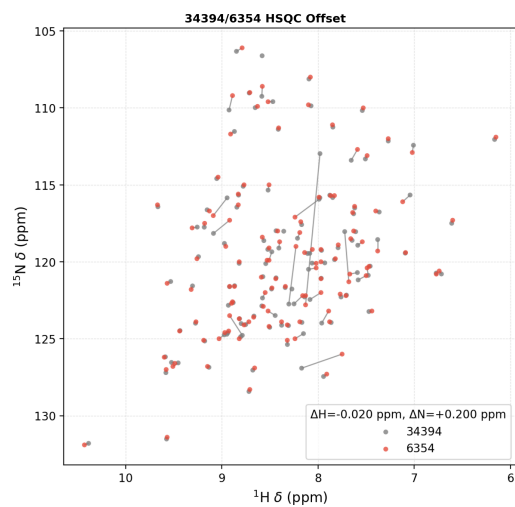
SI Fig. S24C. Apo vs apo control study for BMRB 34000 and 34001. This case involves insulin-like growth factor II (IGF-II). BMRB 34001 is the [S39_PQ]-IGF-II variant and contains a two-residue Pro-Gln insertion after Ser39 relative to wild-type IGF-II (BMRB 34000). 90.6% of the CSPs for this control are below 0.05 ppm.

34394 vs 6354 — apo-apo CSP / HSQC / offset grid



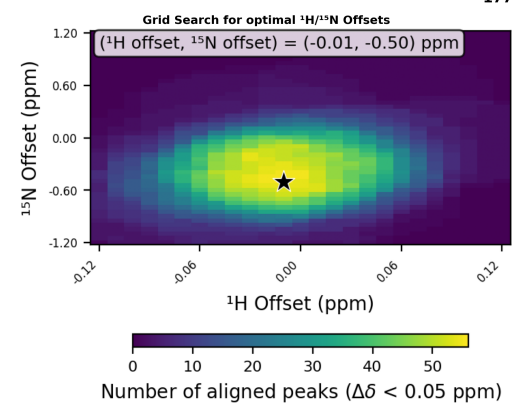
Query BMRB 34394 | PDB N/A
pH=7 T=25 °C pressure=1 atm
Entities: entity 1
Components: (none)

Match BMRB 6354 | PDB N/A
pH=7.2 T=30 °C
Entities: NECAP1 protein
Components: (none)



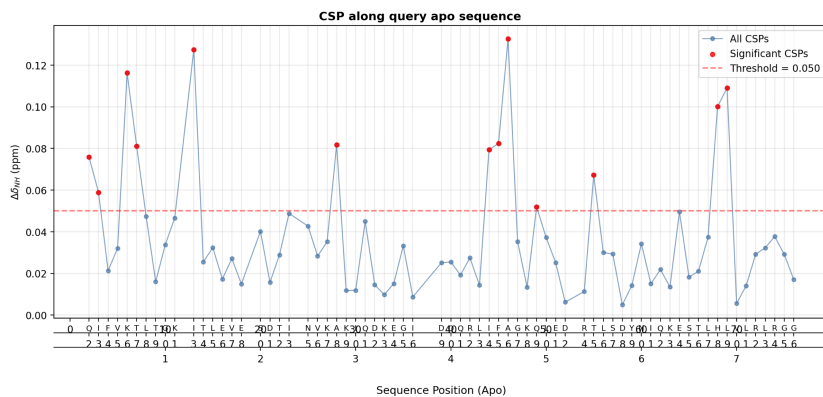
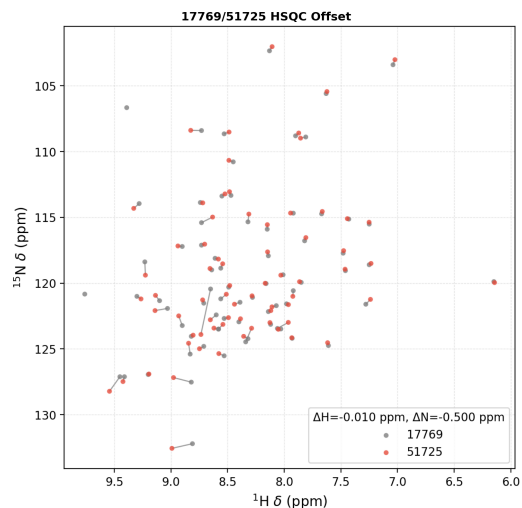
SI Fig. S24D. Apo vs apo control study for BMRB 34394 and 6354. This case involves the NECAP1 Phear domain. BMRB 34394 has a five-residue N-terminal extension (GSPNS) relative to BMRB 6354, plus substitutions T8A and K59I and short internal insertions (Pro28 and Ala31–Ser32) in the 34394 construct. 83.2% of the CSPs for this control are below 0.05 ppm.

17769 vs 51725 — apo-apo CSP / HSQC / offset grid



Query BMRB 17769 | PDB N/A
pH=6.5 T=25 °C pressure=1 atm
Entities: Ubiquitin
Components: (none)

Match BMRB 51725 | PDB N/A
pH=7 T=25 °C pressure=1 atm
Entities: entity_2
Components: Human Ubiquitin T12C 0.7 mM; DTT 0.5 mM; sodium phosphate 50 mM; DSS 0.1 mM

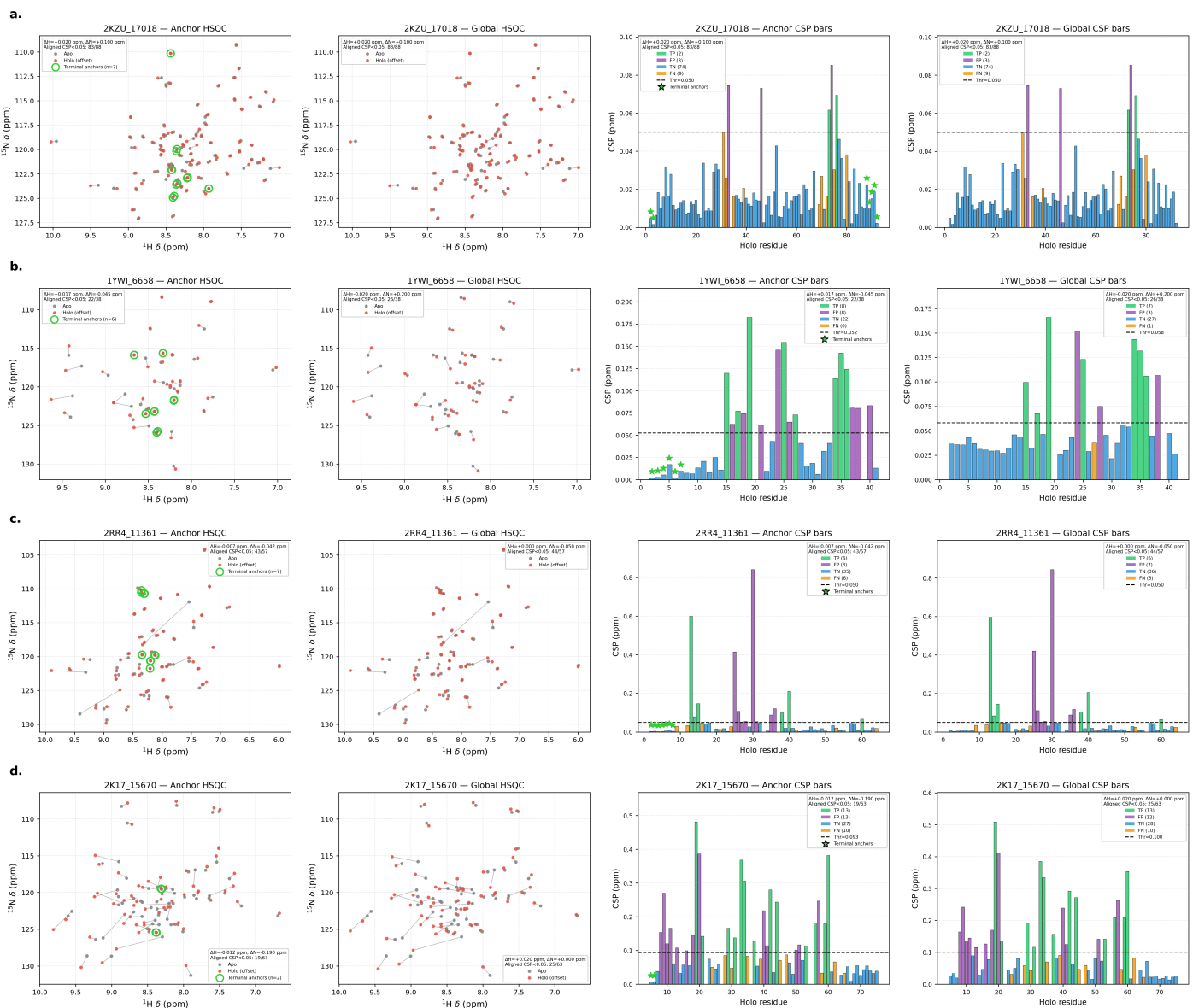


Query / Match CS-list sequence alignment

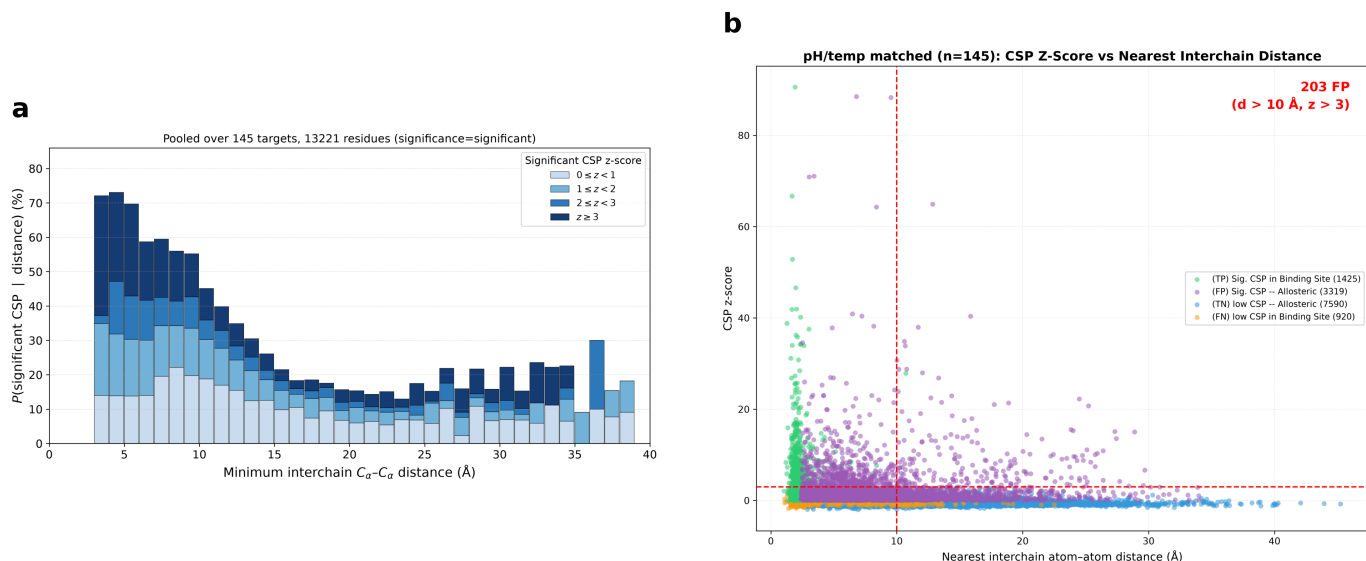
17769	MQIFVKTLLTGK	TITLEVEPSDTIENVKAKIQDKEGIPPDQQRLLIFAGKQL	50
51725	MQIFVKTLLTGK	CITLEVEPSDTIENVKAKIQDKEGIPPDQQRLLIFAGKQL	50
17769	EDGRTLSDYNIQKESTLHLVLRRLGG		76
51725	EDGRTLSDYNIQKESTLHLVLRRLGG		76

SI Fig. S24E. Apo vs apo control study for BMRB 17769 and 51725. This case involves ubiquitin. There is a sequence mismatch at position 12 (T12C; wild-type Thr in BMRB 17769, Cys in BMRB 51725). 81.2% of the CSPs for this control are below 0.05 ppm.

Terminal-anchor vs global offsets: HSQC overlays and CSP classification bars

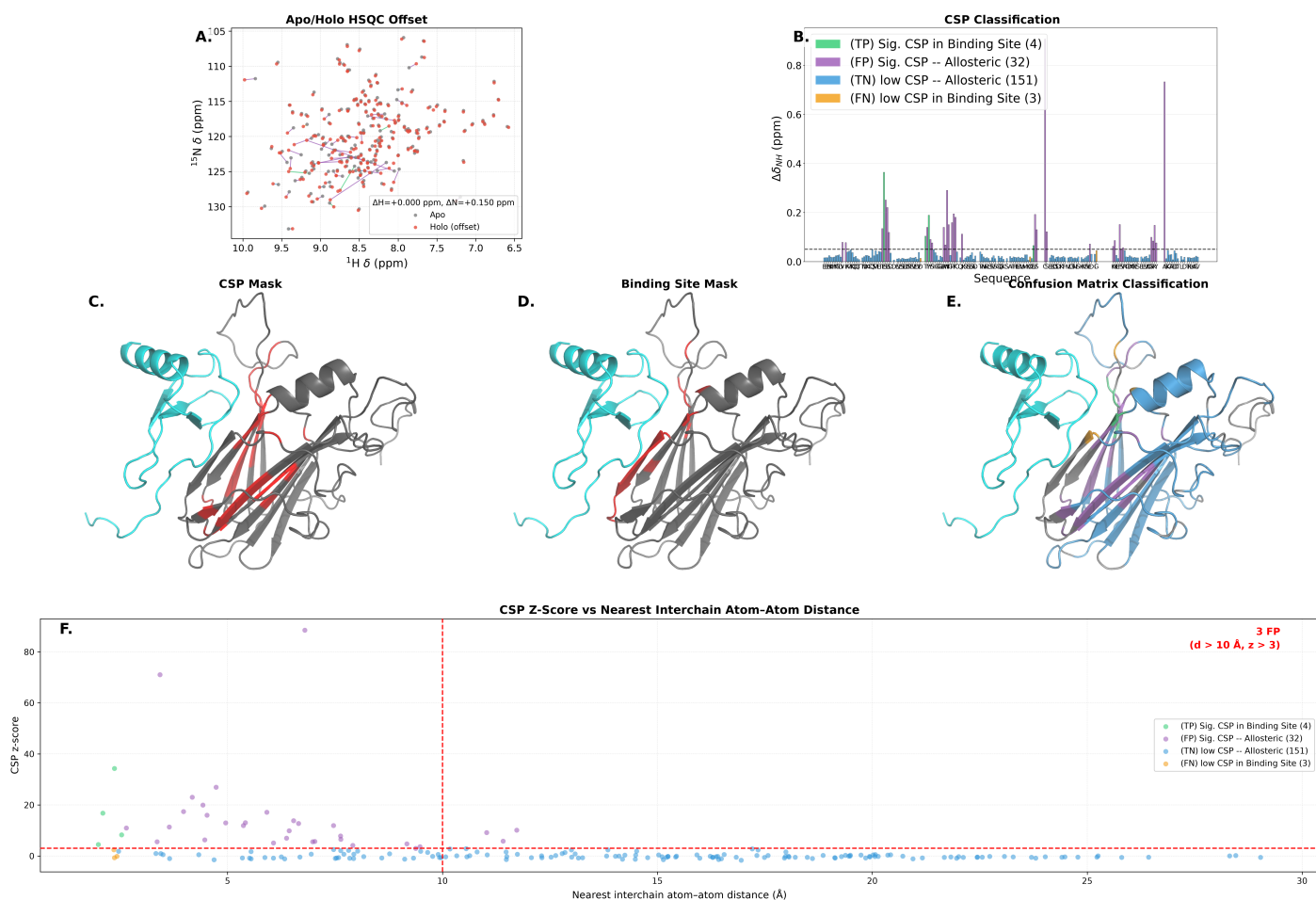


SI Fig. S25. Terminal-anchor vs global offsets: HSQC overlays and CSP classification bars for four targets (2KZU/17018, 1YWI/6658, 2RR4/11361, 2K17/15670). For each target (rows a–d): left pair shows apo (grey) vs offset holo (red) HSQC overlays using terminal-anchor RMSD offsets vs the global CSP-count grid; right pair shows corresponding CSP classification bars (TP/FP/TN/FN) under the primary significance threshold. Green markers denote terminal-anchor residues.



SI Fig. S26. Chemical-shift perturbations as a function of interchain distance. (a) Residue-pooled $P(\text{significant CSP} \mid \text{minimum interchain } C_\alpha\text{-}C_\alpha \text{ distance})$ for buffer-similar systems ($n = 145$ from `CSP_UBQ_ph0.5_temp5C.csv`) under the primary cutoff $\max(\text{cleaned mean}, 0.05 \text{ ppm})$. Bar height is the fraction of residues with a recorded HN CSP that are called significant in each $1 \text{ \AA}$ distance bin; stacks partition those significant CSPs by cleaned-mean z -score ($0 \leq z < 1$, $1 \leq z < 2$, $2 \leq z < 3$, $z \geq 3$). Close to the interface the significant fraction is high and enriched in large z , then declines toward a lower baseline at longer range. (b) Per-residue CSP z -score versus nearest interchain any-atom distance for the same $n = 145$ set (same primary significance definition), colored by confusion-matrix class (TP/FP/TN/FN) relative to the structural binding-site mask. Dashed lines mark $z = 3$ and $d = 10 \text{ \AA}$; the annotation counts false positives with $d > 10 \text{ \AA}$ and $z > 3$.

holo_pdb: 2FIN | apo_pdb: N/A | apo_bmrbs: 6809 | holo_bmrbs: 7024



SI Fig. S27. Case study for holo 2FIN / apo BMRB 6809 (holo BMRB 7024). HSQC overlay, CSP classification bars, CSP / binding-site / confusion structure views, and CSP z versus nearest interchain atom–atom distance. **(F)** Dashed lines mark $z = 3$ and $d = 10$ Å; the annotation counts false positives with $d > 10$ Å and $z > 3$. This plot shows monotonic decay of CSPs with increasing nearest interchain atom–atom distance. Residues in a set of β sheets further away from the ligand with significant shifts have lower CSPs than those closer to the binding site.

$$\text{CSP} = \sqrt{\frac{1}{2}(\Delta\delta_{\text{H}}^2 + (0.14 \Delta\delta_{\text{N}})^2)}$$

SI Equation S1. Combined HN chemical-shift perturbation

$$\text{CSP} = \sqrt{\frac{1}{3}(\Delta\delta_{\text{H}}^2 + (0.14 \Delta\delta_{\text{N}})^2 + (0.3 \Delta\delta_{\text{C}\alpha})^2)}$$

SI Equation S2. Combined HN+C α chemical-shift perturbation

$$\text{CSP} = \sqrt{\frac{1}{2}(\Delta\delta_{\text{H}\alpha}^2 + (0.3 \Delta\delta_{\text{C}\alpha})^2)}$$

SI Equation S3. Combined H α +C α chemical-shift perturbation